get the best results. In general, exponential decay slows down fast at the beginning, and slowly from halfway to the end; figure 17.7 shows a few examples of how the ease of the slowdown depend on the choice of α.

In practice, $\alpha \approx 0.98$ is usually a safe bet for an initial choice (then, you can take it from there and tune it).

17.1.5 Variants

As we have seen, simulated annealing – despite being an indisputably powerful tool, has also some flaws: the more concerning of which is its slow speed in converging, due to its stochastic nature.

And so, like for all algorithms, in time many variants have been studied to remedy its shortcomings, and make it even better.

We already alluded to a trivial modification that we could add: storing the best solution found across all iterations. It might happen, in fact, and especially on large, multi-dimensional problem spaces, that during the initial high-energy phase we serendipitously land in proximity of a global minimum, then move uphill afterwards, and never manage to go back to such a good result again. How likely this is to happen depends on many factors, like the shape of the landscape (the cost function) and if the right configuration for the parameters was found; in any case, remembering the best result ever can be an easy win.

Pushing this consideration a little further, **simulated annealing with restart** stores one or a few of the best